

Present status of and future opportunities for all-perovskite tandem photovoltaics

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All-perovskite tandem solar cells represent the forefront of next-generation photovoltaic technologies, offering a promising pathway to exceeding the Shockley–Queisser efficiency limits of single-junction solar cells while maintaining cost-effectiveness and scalability. However, the transition from laboratory-scale prototypes to commercially viable products faces numerous challenges. Large-area fabrication requires the development of scalable manufacturing techniques while minimizing performance losses compared with laboratory-scale spin coating. Additionally, achieving long-term stability, reliability, efficient integration from cell to module, and high yield during practical deployment remain critical hurdles. Here we address these key aspects, summarize the latest field advancements and highlight strategies to overcome these challenges. By offering insights into the pathway towards reliable, durable and high-performance all-perovskite tandem photovoltaics, we aim to support their deployment in large-scale applications.

The photovoltaic industry has witnessed remarkable progress in recent years, driven by significant cost reductions, efficiency improvements and extended system lifetimes, making it one of the most competitive renewable energy sources. However, as silicon-based photovoltaic technologies mature, their record efficiency of 27.4% for crystalline silicon (c-Si) cells is nearing their practical theoretical limit of ~29.4%¹, leaving limited room for substantial efficiency gains within the c-Si photovoltaic technologies.

To overcome this limitation and further reduce the levelized cost of electricity for photovoltaic electricity generation, a multi-junction architecture approach exceeding the efficiency limit of single-junction devices is highly demanded². A promising pathway towards this goal utilizes all-perovskite tandem solar cells, which leverage the ability of perovskite materials to flexibly tune their bandgaps. This material property allows for two or more subcells with complementary bandgaps

to be integrated into one solar cell device³, as illustrated in Box 1. This technology benefits from a reduction in photon thermalization losses, allowing for record certified power conversion efficiencies (PCEs) reaching beyond 30%, thus outperforming single-junction silicon (27.4%) and perovskite (26.7%) solar cells¹. Moreover, the certified mini-module efficiencies for all-perovskite tandem cells have reached 24.8%, surpassing the 23.2% efficiency of single-junction perovskite modules¹.

Beyond their exceptional efficiency, all-perovskite tandem solar cells benefit from abundant precursor materials, low-energy production and vapour- or solution-processable fabrication, positioning them as a cost-effective alternative to Si-based photovoltaics. Moreover, the capabilities to manufacture tandem cells using low-temperature and scalable processing methods allow for diverse applications, ranging from lightweight and flexible panels to building-integrated photovoltaics. Combined with their potential for further efficiency gains,

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BOX 1

Typical architecture of all-perovskite tandem solar cells

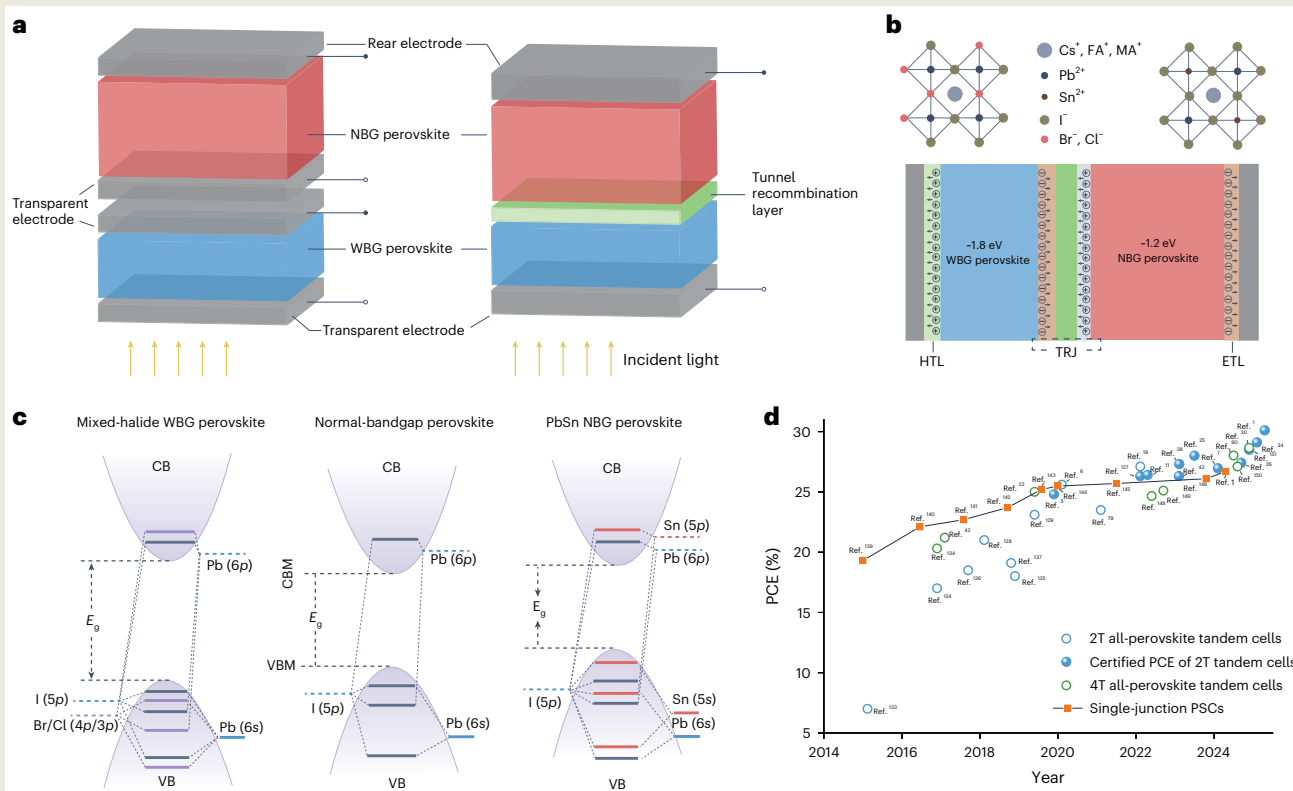
All-perovskite tandem solar cells are classified into two main architectures: two-terminal (2T) and four-terminal (4T). The 4T design employs optical splitting or mechanical stacking, enabling each subcell to operate electrically independently. Although this structure offers resilience towards spectral variations, it incurs a greater complexity and higher production costs compared with 2T tandem solar cells, due to the additional transparent electrodes and interconnection component. In contrast, the 2T architecture integrates both subcells into a monolithic stack with an interconnected TRJ, requiring precise current matching to maximize the power output. This simple design facilitates integration into existing module frameworks and lowers production costs. However, its reliance on current matching can significantly limit the efficiency of the overall tandem if either subcell is considerably less efficient than intended. Additionally, stacking more than ten layers increases the complexity of the fabrication process, potentially reducing the production yield of the device. This results in the need for tight industrial and manufacturing processes.

The TRJ, typically comprising materials such as metal oxides (for example, MoO_x and ITO) or ultrathin metal nanoclusters, enables efficient recombination between electrons (photo-generated from the top cell) and holes (photo-generated from the bottom cell). To realize an efficient performance, the TRJ needs high vertical conductivity and minimized lateral conductivity to prevent shunting.

Ideally, the tandem device's V_{OC} equals that of each subcell with minimal fill factor loss¹³⁰. Additionally, the TRJ requires high optical transmittance to minimize reflection and parasitic absorption losses, ensuring efficient light harvesting.

The tunability of the perovskite's bandgap allows for each of the subcells to achieve a current match in a 2T tandem architecture. In ABX_3 -type perovskites, the A site is typically occupied by a mixture of cations such as Cs^+ , FA^+ and MA^+ . Compared with adjustments at the A site, the bandgap is controlled by hybridization of the B-site cationic orbitals with the X-site anionic orbitals, and modifications at these sites can achieve the desired bandgap¹³¹. For 2T tandem solar cells, WBG (~1.8 eV) perovskites are achieved by replacing I^- with smaller halide ions, such as Br^- or Cl^- at the X site, while Sn doping at the B site produces the NBG (down to ~1.2 eV) perovskites, expanding the utilization of the near-infrared solar spectrum efficiently¹³².

Research on 2T tandem architecture has gained considerable attraction, with rapid performance advancements^{5–7,11,19,25,26,30,34,38,42,127–129,133–138}. The certified efficiencies for 2T tandem cells have now exceeded 30%¹, far surpassing those of single-junction PSCs (26.7%)^{1,139–147} and 4T tandem cells (28.6%)^{22,90,129,134,148–151}. Given the rapid progress and significant potential for commercialization, this Review focuses on the advancements and challenges associated with 2T all-perovskite tandem solar cells and modules.



Box Fig. 1 **a**, Classification of all-perovskite tandem devices into 2T (right) and 4T (left) configurations. **b**, Perovskite materials and working principles of 2T all-perovskite tandem solar cells. $V_{\text{sum}} = V_{\text{WBG}} + V_{\text{NBG}}$. **c**, Schematic of how ion doping modifies the bandgap and energy levels of perovskite materials. **d**, Comparison of the efficiency progress of 2T and 4T all-perovskite tandem solar cells against single-junction counterparts^{1,5–7,11,19,22,25,26,30,34,38,42,90,127–129,133–151}. CB, conduction band; CBM, conduction band minimum; E_g , bandgap energy (that is, the minimum energy needed for an electron to jump from the valence band to the conduction band); VB, valence band; VBM, valence band maximum.

these unique properties position all-perovskite tandem cells as a pivotal technology for large-scale solar energy deployment. However, transitioning from laboratory prototype cells to commercially viable module products requires overcoming critical scalability, stability and real-world reliability challenges.

This Review provides a comprehensive overview of recent progress in all-perovskite tandem solar cells, focusing on strategies to improve their efficiency and address challenges in stability and scalability. It examines the mechanisms driving efficiency improvements, emphasizing the unique features and critical challenges of the positive–intrinsic–negative (p–i–n) architecture, also known as the inverted architecture of the wide-bandgap (WBG) and narrow-bandgap (NBG) subcells. It further explores material and structural factors influencing tandem stability, with a focus on emerging methodologies to improve durability. Next, the transition from small-area devices to large-area modules is discussed, with bottlenecks in large-area processing highlighted. Finally, key directions for future research are suggested and a roadmap to the advancement of all-perovskite tandem solar cells towards practical applications is offered.

Strategies to exceed the single-junction efficiency limit

The recent efficiency advancements in all-perovskite tandem solar cells have been mainly driven by three key factors: innovations in tunnel recombination junctions (TRJs) (Boxes 1 and 2) and improvements in both WBG and NBG subcells (Fig. 1a). These breakthroughs were primarily rooted in an enhanced understanding of the p–i–n architecture of perovskite solar cells (PSCs), which has enabled significant efficiency gains in both subcells. Key factors contributing to these gains encompass: a significant reduction in non-radiative defects in the absorber layers, optimized surface passivation treatments and improved charge-carrier transport layers to enhance the electron extraction, hole extraction, electron transport or hole transport properties while suppressing surface recombination at that interface (Fig. 1b,c).

Unique issues in NBG perovskite subcells

NBG perovskites present distinct challenges compared with Pb-based perovskites, including the intrinsic susceptibility of Sn^{2+} to oxidation, rapid crystallization leading to low-quality films, brittle contacts at the interface between poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT:PSS) and the perovskite and residual solvents within the perovskite films, especially at buried interfaces.

The oxidation of Sn^{2+} to Sn^{4+} is visually evident with a colour change in the precursor solution and can also occur through more subtle mechanisms, such as solvent-mediated and charge-induced oxidation, resulting in additional Sn vacancy defects⁴ (Fig. 1d). This process occurs even in trace oxygen and accelerates at 65–85 °C due to the high reactivity and mobility of Sn^{2+} , making its retention intrinsically challenging. To address these issues, several redox-based strategies have been developed. Comproportionation, utilizing metallic Sn to reduce Sn^{4+} to Sn^{2+} in the precursor solution, extends the charge-carrier diffusion length in Pb–Sn perovskites and achieves PCEs exceeding 21%⁵ (Fig. 1e). Amphiphilic antioxidants with surface-anchoring properties, such as formamidine sulfonic acid, enhance film uniformity, yielding a PCE of 21.7%⁶. Additionally, electron-withdrawing ligands such as chloromethyl phosphonic acid mitigate Sn oxidation⁷, reducing the defect density and achieving efficiencies of 21.65% with an open-circuit voltage (V_{oc}) of 0.852 V. Recently, multifunctional additives, such as 2-amino-3-mercaptopropanoic acid hydrochloride and thiomalic acid^{8,9}, which contain functional groups such as a thiol group (–SH), carboxylic acid group (–COOH) and ammonium group (– NH_3^+), have been designed to prevent Sn^{4+} formation and have pushed the PCE of NBG subcells to around 23%.

Driven by the high Lewis acidity of Sn^{2+} and its fast coordination with organic components, the rapid crystallization of NBG perovskites

poses significant challenges in fabricating high-quality films. This process often produces small grains with high-density grain boundaries that might act as non-radiative recombination centres and trap sites (Fig. 1f). Additionally, Sn alloying induces grain contraction, leading to residual strain that exacerbates defect formation and hinders charge extraction¹⁰. The weak Sn–I bonding compared with that of Pb–I further causes organic cation depletion at the film surface, increasing the trap density. Key innovations have focused on optimizing charge transport and defect passivation to address these issues. Incorporating 4-trifluoromethyl-phenylammonium into the precursor solution significantly extended the carrier diffusion length to over 5 μm , enabling the fabrication of thicker NBG perovskite layers (up to 1.2 μm)¹¹. This advancement improved the short-circuit current density (J_{sc}) and led to efficiencies exceeding 22% for NBG subcells, contributing to a certified tandem efficiency of 26.4% that surpassed those of single-junction PSCs. It is noteworthy that, to ensure adequate light absorption, the thickness of the NBG perovskite layer typically needs to exceed 1 μm . However, due to incomplete solvent quenching, this can result in residual solvents at buried interfaces, which may compromise device stability.

Another noteworthy approach involves introducing quasi-two-dimensional (quasi-2D) structures, such as $(\text{PEA})_2\text{GAPb}_2\text{I}_7$, by using mixed bulky organic cations (for example, phenethylammonium (PEA^+) and guanidinium (Gua^+))¹². These structures effectively passivate defects and enhance the structural and optoelectronic properties of 1.25-eV Pb–Sn perovskites. These advancements addressed crystallization-induced defects and charge transport limitations, enabling NBG subcell efficiencies exceeding 22% and tandem cell efficiencies up to 25.5%.

Unique issues in WBG perovskites

For WBG perovskite subcells, the experimentally reported V_{oc} values have consistently fallen short of theoretical maximums compared with normal-bandgap devices¹³. Recent advancements in understanding the underlying causes of V_{oc} losses have shifted the focus beyond just phase segregation due to high Br content. Although phase segregation remains a critical issue for long-term stability (as discussed below), the increased V_{oc} in WBG perovskite films is often linked to deep trap states induced by halide interstitials¹⁴. A higher Br content can amplify the formation of transient hole traps, adversely impacting quasi-Fermi level splitting and thus photoluminescence quantum yields¹⁵ (Fig. 1c). Density functional theory studies have shown that halide interstitials are thermodynamically stable and contribute significantly to the formation of deep trap states¹⁶. Advanced characterization techniques, such as photoemission electron microscopy and multimodal operando microscopy have been developed to enable nanoscale mapping and spatial correlation of charge transport losses, recombination losses and chemical composition, providing critical insights into the spatial heterogeneity within perovskite films and the origins of performance limitations^{17,18}. Non-radiative recombination pathways are particularly pronounced at grain boundaries and surfaces in perovskite films, driven by the complex crystallization kinetics of mixed cations and halides.

Addressing the above challenges requires strategies to either homogenize the crystallization process or effectively passivate surface and bulk defects. Notably, gas quenching has been highly effective for producing components with a higher caesium (Cs^+) content, resulting in improved structural quality and higher V_{oc} values^{19–21} (Fig. 1g). A promising approach involves a film-side-down close-space annealing process, in which solvent release is precisely controlled, resulting in high-quality perovskite absorbers with micrometre-scale grains and prolonged carrier lifetimes²². For WBG perovskites, additive strategies, such as incorporating 2D passivation, chloride- or pseudohalide-based salts and multifunctional small molecules²³, have been employed to regulate film morphology and reduce defect states. Despite these advancements, the inherent crystalline heterogeneity of WBG materials

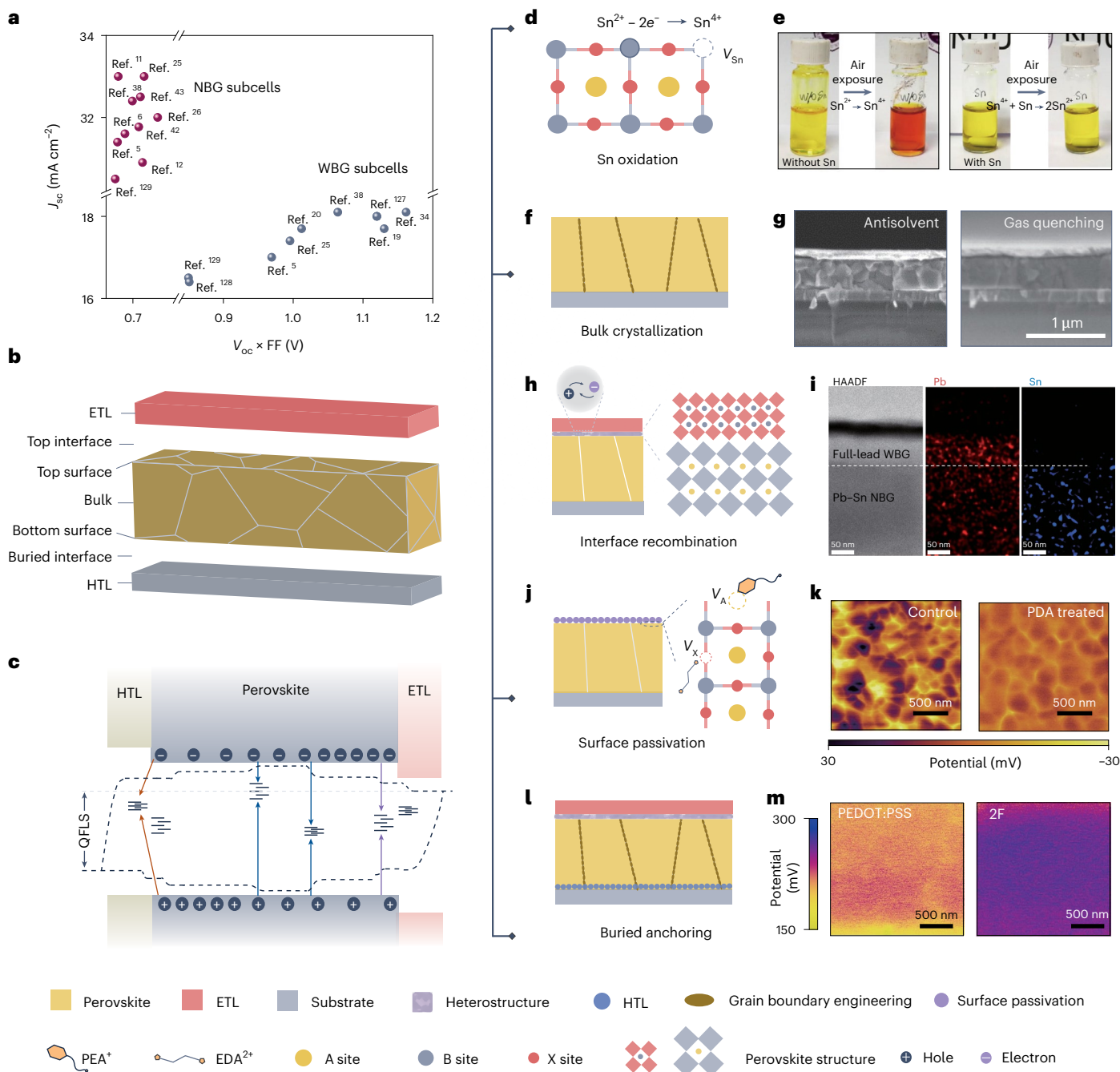


Fig. 1 | Strategies to enhance efficiency in WBG and NBG subcells. **a**, Statistical performance parameters of WBG and NBG perovskite subcells in tandem configuration^{5,6,11,12,19,20,25,26,34,38,42,127–129}. **b**, Schematic of the p-i-n architecture of PSCs. **c**, Non-radiative recombination pathways in devices. The black circles with plus and minus signs represent holes and electrons, respectively. The horizontal dashed lines indicate split Fermi levels. The staggered short lines denote non-radiative recombination centres. The orange, blue and purple arrows indicate non-radiative pathways at the HTL interface, perovskite bulk and ETL interface, respectively. **d–m**, Sources of performance losses in PSCs with p-i-n architecture. **d**, Schematic illustrating the issue of Sn^{2+} being oxidized to Sn^{4+} in NBG films by losing two electrons. **e**, Photographs showing the ease of oxidation of Sn^{2+} to Sn^{4+} under an atmosphere of ambient air and the facile reduction of Sn^{4+} to Sn^{2+} by metallic Sn powders. **f**, Schematic of perovskite bulk crystallization regulation and grain boundary passivation. **g**, Cross-section scanning electron microscopy images of WBG perovskite films made with the anti-solvent and gas

quenching methods. **h**, Schematic of heterojunction formation at the perovskite interface. **i**, Cross-sectional high-resolution transmission electron microscopy image (HAADF) and corresponding energy-dispersive X-ray mapping of Pb-Sn PSCs with a bilayer perovskite heterojunction. **j**, Schematic of molecular passivation on the perovskite surface. **k**, Kelvin probe force microscopy images of control and 1,3-propane-diammonium iodide (PDA)-treated WBG films. **l**, Schematic of HTL improvement and buried passivation anchoring in devices. **m**, Kelvin probe force microscopy images of different NBG perovskite films deposited on PEDOT:PSS and MPA2FPh-BT-BA (2F). EDA, ethylenediammonium; HAADF, high-angle annular dark-field imaging; FF, fill factor; QFLS, quasi-Fermi level splitting. Panels reproduced with permission from: **e**, ref. 5, Springer Nature Limited; **g**, ref. 19, AAAS; **m**, ref. 38, Springer Nature Limited. Panels adapted with permission from: **k**, ref. 127, Springer Nature Limited; **i**, ref. 25, Springer Nature Limited.

still results in more significant losses compared with the highly crystalline quality of FAPbI₃ films, underscoring the need for further exploration and refinement.

Interface engineering in p–i–n structure

Following the reduction of bulk defects, strategies have increasingly focused on reducing non-radiative recombination between the perovskite and hole transport layer (HTL) or electron transport layer (ETL) interfaces. High-density traps at perovskite heterojunction interfaces, often formed by unpassivated dangling bonds, are among the primary causes of recombination losses²⁴. By tailoring the terminal groups at the perovskite interface, surface modification effectively reduces defects and further improves charge extraction.

Perovskite/ETL interface

Various surface modification techniques have substantially improved the performance of p–i–n PSCs by addressing two critical aspects: (1) forming heterojunctions to induce n-type properties in the top surface region, facilitating efficient charge transfer (Fig. 1h); and (2) reducing surface defects to minimize recombination losses (Fig. 1j). For instance, strategies to create a type-II band alignment at the Pb–Sn perovskite/ETL interface have been employed to suppress the undesired p-type doping concentration, reduce the defect-induced surface recombination velocity and improve electron extraction²⁵ (Fig. 1i). These approaches have enabled Pb–Sn PSCs with 1.2- μm -thick absorbers and have thus achieved a high PCE of 23.8%, with a V_{oc} of 0.873 V and a fill factor of 82.6%²⁵. Further research has revealed that mixed Pb–Sn perovskite films often exhibit compositional gradients with excess Sn at the top surface, leading to oxidation and accelerated recombination rates²⁶. Diamines have been used to chelate Sn atoms to address this issue, achieving a more balanced Sn stoichiometry. This strategy forms a low-resistance barrier layer that passivates defects, enabling efficiencies of up to 23.9% in NBG devices.

Optimizing the perovskite/ETL interface has also improved V_{oc} and fill factor values in WBG subcells for tandem architecture. Three main strategies have been developed: (1) anchoring functional materials on the perovskite surface; (2) constructing low-dimensional materials on top of the perovskite absorber; or (3) constructing heterostructured materials on top of the perovskite absorber. Organic salts such as guanidinium bromide, phenethylammonium iodide, butylammonium bromide and 4-(trifluoromethyl)phenylethylammonium are widely used as passivation to stabilize the perovskite surface and reduce non-radiative recombination losses^{27–29}. An effective surface reconstruction strategy was developed to fabricate a high-quality NBG perovskite surface with a close-to-ideal stoichiometric ratio by using 1,4-butanediamine (to minimize Sn⁴⁺-related defects) and ethylenediammonium diiodide (to passivate organic cation and halide vacancy defects)³⁰. Co-modifiers such as diammonium combined with methylthio or phenylethylammonium molecules have also demonstrated selective passivation for defect states, reducing trap-assisted recombination in both WBG and NBG perovskites^{31,32} (Fig. 1k). Furthermore, diammonium has been reported to enhance the orientation of [6,6]-phenyl-C₆₁-butyric acid methyl ester, thereby minimizing non-radiative recombination at the perovskite/ETL interface³³. A recent strategy using a mixture of methylammonium iodide (MAI) and phenethylammonium iodide in the anti-solvent enabled the fabrication of WBG perovskites with enhanced (100) plane orientation, achieving a certified tandem PCE of over 29%³⁴. This method promotes 2D ($n = 1$) layered perovskite formation, oriented nucleation and downward crystal growth, reducing non-radiative recombination in the bulk and interfaces. In addition to organic interface engineering, ultrathin dipole layers, such as MgF₂ or LiF, inserted at the perovskite/C₆₀ interface mitigate non-radiative recombination and enhance device performance³⁵.

Perovskite/HTL interface

The buried perovskite/HTL interface poses a significant challenge for p–i–n PSCs. The high density of deep trap states at this interface has been revealed through mechanical and chemical delamination methods^{36,37} (Fig. 1l). Advances in HTLs (for example, replacing the traditional PEDOT:PSS with multifunctional materials such as MPA2FPh-BT-BA; Fig. 1m) have led to improved energy-level alignment, regulated perovskite growth and effective defect passivation³⁸. Specifically, MPA2FPh-BT-BA utilizes a benzoic acid anchoring group to replace the traditional 2-cyanoacrylic acid, improving molecular stacking and face-on orientation on indium tin oxide (ITO) surfaces, whereas the introduction of fluorine atoms deepens the highest occupied molecular orbital level. Another designed self-assembled monolayer (SAM) with oligoether side chains was custom introduced on the benzothiadiazole unit by the same research group, pushing the efficiency of NBG subcells to 23.54%³⁹. Since 2022, SAMs anchored on transparent conductive oxide (TCO) or nickel oxide (NiO_x) as HTLs have emerged as mainstream HTLs in WBG subcells⁴⁰, replacing traditional conductive polymers such as poly[bis(4-phenyl)(2,4,6-trimethyl-phenyl)amine] and N4,N4'-di(naphthalen-1-yl)-N4,N4'-bis(4-vinylphenyl)biphenyl-4,4'-diamine. Early studies on SAMs, such as (2-(9H-carbazol-9-yl)ethyl)phosphonic acid and Me-[4-(9H-carbazol-9-yl)butyl]phosphonic acid, demonstrated enhanced hole extraction and reduced interface traps, significantly reducing voltage losses in WBG devices. Subsequent development of SAMs, including (4-(7H-dibenzo[c,g]carbazol-7-yl)butyl)phosphonic acid, has further optimized the passivation efficacy, energy-level alignment and surface wettability, resulting in superior device performance⁴¹.

Despite these advancements, the passivation of the buried interface has received less attention than the perovskite/ETL interface, mainly due to the solubility of passivation materials in perovskite precursor solvents. A doping strategy using chiral aspartate hydrochloride at the PEDOT interface was developed to address this limitation. This approach forms intermolecular hydrogen bonds throughout the perovskite film, enabling an efficiency of 22.46% in Pb–Sn PSCs⁴². Beyond doping HTLs with pre-embedded passivation molecules, glycine hydrochloride additive was utilized to passivate the buried perovskite/ETL interface and improve the quality of the underlying perovskite layer, as the large-size GlyH⁺ cations preferentially settle to the bottom during crystallization^{41,43,44}. From another perspective, a vapour solution two-step method appears more suitable for bottom-surface passivation. However, this approach currently yields lower-quality perovskite films than the one-step spin-coating method, highlighting a need for further optimization.

Operational stability of all-perovskite tandem solar cells

As the efficiency of all-perovskite tandem solar cells surpasses the 30% benchmark, device stability becomes a critical issue, which is currently hindering their commercial potential. In recent years, significant progress has been made in enhancing the operational lifetimes of all-perovskite tandem solar cells (Fig. 2) through material design, crystallization regulation, defect passivation and device structure optimization. Among these advancements, the most notable breakthroughs include mitigating halide segregation in WBG perovskites, optimizing the composition and structure of NBG perovskites and developing robust interconnection layers. This section highlights these key developments and their role in advancing device stability.

Light-induced halide segregation in WBG perovskites

WBG perovskites are more susceptible to halide segregation under continuous illumination, driven by photoinduced polarons and halogen ion migration^{45,46}. This segregation forms iodide (I)- and bromide (Br)-rich regions, where photocarriers become trapped in low-bandgap I-rich

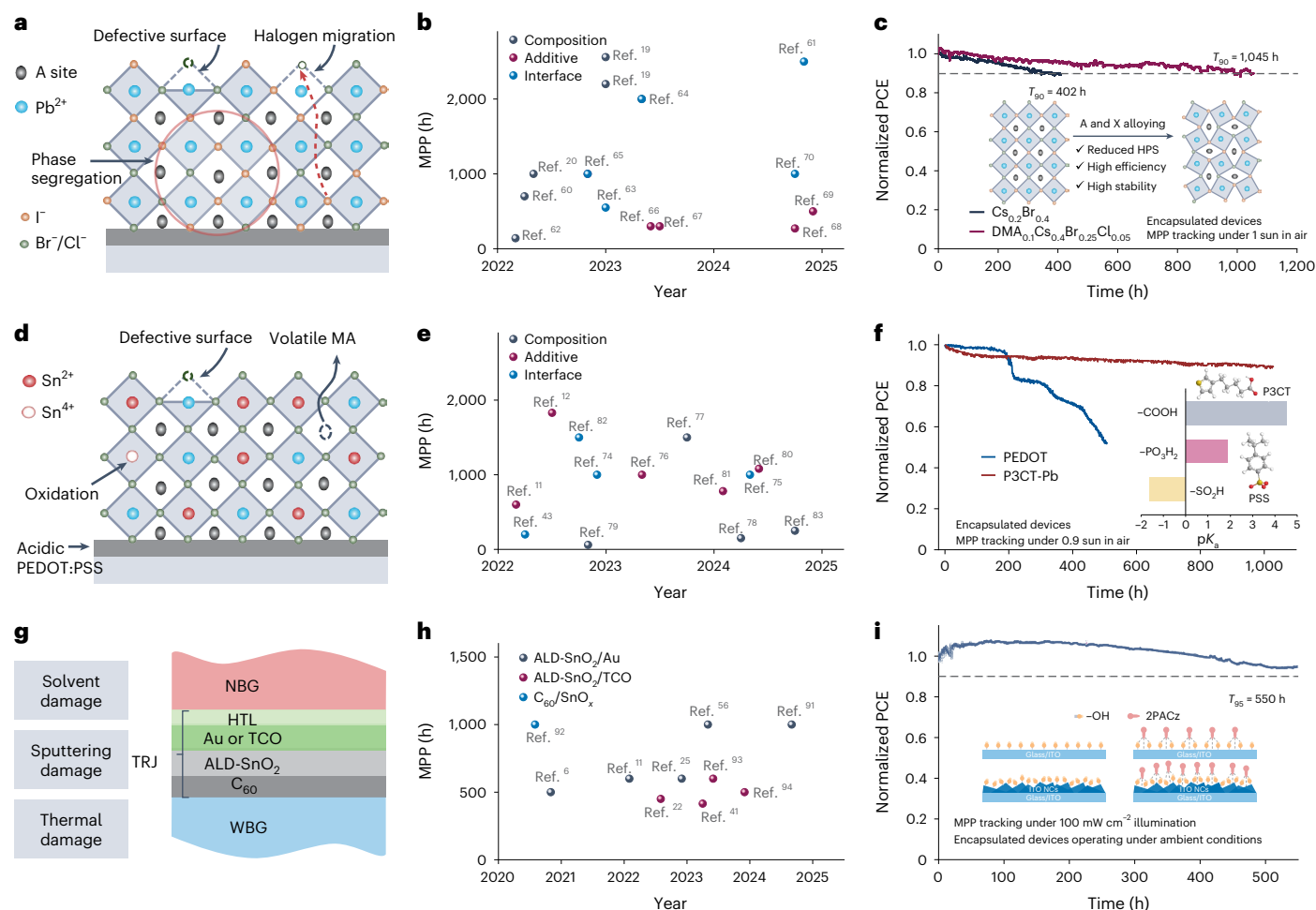


Fig. 2 | Operational stability of subcells and tandem photovoltaics. **a**, Stability issues of WBG perovskites. **b**, Summarized MPP values of WBG subcells^{19,20,60–70}. **c**, MPP stability of WBG solar cells with A- and X-site alloying. The inset shows the effects of A- and X-site ion engineering on the stability of WBG perovskite films. **d**, Summarized stability issues of NBG perovskites. MA, methylamine. **e**, Summarized MPP values of NBG subcells^{11,12,43,70,74–83}. **f**, MPP stability of NBG solar cells with P3CT HTLs. The inset shows the calculated pK_a values of functional groups attached to the methyl group in P3CT. **g**, Stability issues of

all-perovskite tandem solar cells originating from TRJs. **h**, Summarized MPP values of all-perovskite PSCs with different TRJs^{6,11,22,25,41,56,91–94}. **i**, MPP stability of the state-of-the-art all-perovskite tandem solar cells with a TRJ structure of $\text{SnO}_2/\text{NC-ITO/SAMs}$. The inset shows a schematic of bonding between SAMs and ITO or ITO nanocrystals (NCs). PACz, 9H-carbazol-9-yl(ethyl)phosphonic acid. Panels reproduced with permission from: **c**, ref. 20, Wiley; **f**, ref. 91, Cell Press. Panel **i** adapted with permission from ref. 94, Wiley.

domains, increasing non-radiative recombination loss. The phenomenon is further exacerbated by lattice defects, grain boundaries and residual stress (Fig. 2a), underscoring the need for thermodynamically stable compositions and high-quality films.

Bulk strategies to address halide segregation focus on two main approaches: adjusting compositions to increase the activation energy for halide migration and reducing defects to suppress ion migration pathways (Fig. 2b). Recent studies have highlighted that halide segregation is not solely determined by Br content; the A-site cation also plays a significant role in the overall lattice structure^{47,48}. Although incorporating Cs enhances stability, it introduces deep-level defects due to its size mismatch with formamidinium (FA^+). To mitigate this, larger A-site cations (for example, dimethylammonium (DMA^+), Gua⁺ and ethylammonium (EA^+)) have been introduced to reduce lattice strain and defect densities^{49,50}. At the X site, chloride (Cl^-) doping has been shown to increase the halide migration barrier by alleviating lattice microstrain, thereby stabilizing the lattice framework⁵¹. A representative composition, $\text{Cs}_{0.4}\text{FA}_{0.5}\text{DMA}_{0.1}\text{Pb}(\text{I}_{0.72}\text{Br}_{0.23}\text{Cl}_{0.05})_3$, achieves a bandgap of 1.8 eV with an optimized PCE and improved photostability (it has a T_{90} value—representing the time required to degrade to 90% of the initial efficiency—of over 1,000 h)²⁰ (Fig. 2c).

Defects in WBG perovskite films act as low-energy pathways for ion migration, significantly reducing operational lifetimes. Theoretical studies indicate that phase segregation often originates at grain boundaries, where defect vacancies are concentrated⁵². Crystallization regulation strategies to address these issues are discussed above. Here we highlight the critical role of interfaces in governing the photostability of WBG perovskite devices. Interfacial defects not only increase non-radiative recombination, thereby reducing efficiency, but also lead to carrier accumulation within the perovskite layer, often resulting in irreversible degradation^{53,54}. Studies have shown that under short-circuit conditions, where carrier accumulation is minimized, phase segregation in WBG perovskites is significantly suppressed⁵⁵. In contrast, under open-circuit conditions, carrier accumulation can induce the formation of polarons, which amplify phase segregation and accelerate degradation⁵⁶. This phenomenon explains the disparity between experimental observations of phase segregation occurring within minutes in thin films and the T_{90} operational lifetimes of WBG devices, which can extend to several hundred hours under real-world conditions.

As the dominant HTLs in WBG subcells, SAMs can be further optimized by mixing multiple types to improve energy-level alignment,

enhance carrier collection and ultimately boost the stability of WBG perovskite devices. However, their single-molecule anchoring mechanism can result in localized microscopic non-uniformity and susceptibility to solvent-induced detachment^{57,58}. Moreover, their long-term thermal stability remains uncertain, with potential risks of substrate delamination or desorption under prolonged stress. It is also worth exploring that the intrinsic stability of transport layers cannot be equated to their stability when integrated into devices. For example, although NiO_x exhibits excellent intrinsic stability, its interfaces with perovskites often suffer from chemical reactions and significant defects, making them a primary source of degradation⁵⁹. Striking a balance between these factors remains a key challenge for future research^{60–70}.

Incorporating thin, low-dimensional perovskite layers into top interfaces has enhanced stability, leveraging the hydrophobic nature of large organic cations and their higher ion migration barriers. However, conventional spin-coating methods often yield a 2D layer with $n = 1$ due to a solvent-driven surface reconstruction displacement reaction. Although $n = 1$ layers at the upper interface facilitate hole transport in n -i-p configurations, their energy-level alignment impedes electron transport at the C_{60} interface in p-i-n architectures, diminishing the stability gains of the 2D layer. To address this, several strategies have been developed to form higher- n 2D layers with more appropriate energy-level alignment, such as solvent modulation during passivation, avoiding thermal annealing and in situ formation via evaporated PbI_2 on the surface^{71–73}. These approaches have improved energy-level alignment and enhanced electron transport while optimizing stability and device performance, enabling the T_{90} of WBG subcells to reach up to 1,700 h (ref. 56).

Stability issues in NBG perovskites

The oxidation of Sn^{2+} and rapid crystallization leading to high defect states (Fig. 2d) are strongly linked to stability, but as they are discussed in the section ‘Strategies to exceed the single-junction efficiency limit’, they are not discussed extensively here^{74–76}. Although methylammonium (MA^+) ions have been shown to enhance stability under specific scenarios, their inclusion at levels exceeding 30% in high-efficiency NBG compositions poses considerable stability risks^{77–83}. The volatility, hygroscopicity and thermal instability of MA^+ significantly compromise device durability, particularly under elevated temperatures, such as the 85 °C conditions specified in International Electrotechnical Commission standards. To address these issues, pure FA-based Pb–Sn perovskites were developed, achieving efficiencies of 21.0% in single-junction cells and 26.3% in all-perovskite tandem cells, with a T_{90} thermal stability of 212 h at 85 °C under dark conditions⁷⁹. Incorporating Cs into FA-based systems further enhanced thermal robustness, giving rise to double- and triple-cation compositions, such as FACs and FARbCs ⁸⁴. These systems demonstrated improved resistance to light and humidity, with stabilized efficiencies and significant gains in long-term operational stability. For instance, advanced RbCsFA compositions have achieved a record efficiency of 24.13% and thermal stability (T_{90}) of 795 h (ref. 85). All-inorganic $\text{CsPb}_{0.4}\text{Sn}_{0.6}\text{I}_3$ perovskites have also shown promise, offering superior thermal stability, with T_{80} exceeding 1,510 h at 65 °C and 800 h at 85 °C, despite a relatively lower efficiency of 17.4%⁸⁶. These results highlight the potential for further optimization of inorganic compositions.

The acidity of PEDOT:PSS, a commonly used HTL, negatively impacts device stability by damaging the brittle interface and creating detrimental recombination centres. Modifications to PEDOT:PSS, such as p-guanidinobenzonitrile hydrochloride treatment, have led to an improvement in performance, achieving a PCE of 23.3% with a T_{80} of 300 h (ref. 87). Exploring alternatives to PEDOT:PSS is a valuable direction. Additives such as multifunctional ammonium sulfide⁸⁸, oxalic acid⁸⁹ and acetylcholine chloride have been used to modify PEDOT:PSS⁹⁰, achieving PCEs over 24% in NBG subcells. These strategies

have also demonstrated efficiencies of ~28% with a T_{80} of 500 h under maximum power point (MPP) conditions for all-perovskite tandem cells. SAMs are another promising approach for Pb–Sn devices, yielding PCEs of up to 23.46% with a T_{88} of 590 h at 65 °C⁴¹. Impressively, GeI_2 -modified NiO_x achieved a PCE of 23.34% with storage stability exceeding 4,320 h in N_2 (ref. 83). A novel HTL, P3CT, with a higher $\text{p}K_a$, overcame interface degradation and enabled tandem cells to retain 96% of their initial PCE after >1,000 h of MPP testing under ambient conditions (Fig. 2e,f), representing high stability for this device type⁹¹.

Stability issues from interconnection layers

Although the SnO_2 /ultrathin Au/PEDOT:PSS configuration is the mainstream TRJ structure for all-perovskite tandem cells, concerns about the scalability of this design arise due to the potential interfacial diffusion of Au clusters during prolonged operation (Fig. 2g,h). To address these concerns, various interconnection layer innovations have been proposed to enhance stability^{92–94}. One promising alternative is the SnO_2 /NC-ITO/SAMs structure, which combines high transparency and excellent electrical conductivity. The ITO nanocrystal (NC-ITO) layer was fabricated using a low-temperature solution-processing method, reducing damage to the underlying subcells and demonstrating superior stability with a T_{95} stability of 550 h (ref. 94) (Fig. 2i). Another widely used TRJ configuration is the SnO_2 /sputtered TCO/PEDOT structure, where sputtered TCO films such as ITO and indium zinc oxide provide high transparency and conductivity^{41,93}. In perovskite–organic tandem cells, a sputtered indium zinc oxide layer sandwiched between bathocuproine or SnO_x and MoO_x has achieved a record PCE of 24% by minimizing optical and electrical losses^{95,96}. However, the sputtering process, particularly at high temperatures or with high-energy particles, risks irreversible damage to the underlying perovskite subcell and limits its adoption in high-efficiency all-perovskite tandem devices.

A novel approach employs an n- SnO_x /p- SnO_{2-x} TRJ, leveraging the bipolar properties of SnO_x for efficient electron and hole transport⁹¹. The simplified C_{60} / SnO_{2-x} interconnection layer further improves the efficiency and photostability of two-terminal (2T) all-perovskite tandem cells by reducing the reliance on complex TRJ structures and minimizing sputtering-induced damage. However, the need for precise control of pulse and reaction times to achieve the optimal Sn:O ratio has hindered its widespread application.

Perspective for stability research

Current stability research for tandem devices lags behind that for single-junction cells, with most studies limited to International Summit on Organic and Hybrid Photovoltaics Stability (ISOS) protocols⁹⁷. Although both NBG and WBG subcells generally adhere to established standards, the absence of specialized stability protocols for all-perovskite tandem solar cells presents a critical issue. Current ISOS protocols, designed for conventional single-junction devices, use single-stress tests (such as light, heat or moisture alone), failing to mimic the multistress conditions (such as simultaneous light, heat and electrical bias) found under real-world environments. Moreover, achieving tandem efficiency crucially depends on precise current matching, yet there is inconsistency in the light sources used in studies (for example, a two-lamp high-spectral-match solar simulator versus a white light-emitting diode), leading to skewed results in interlaboratory stability and performance comparisons.

Module-scale tandem cells face their own set of challenges, such as interlayer delamination due to thermomechanical stresses, halogen outdiffusion across subcell interfaces and encapsulation failures under combined humidity–thermal cycling. The current ISOS protocols, such as ISOS-T for thermal ageing, do not sufficiently cover tandem cell-specific degradation modes, including voltage-mismatch-induced leakage currents or subcell-selective degradation under partial shading. Several urgent standardization efforts are necessary: (1) material- or subcell-level tests (for example, resistance to ultraviolet radiation for

WBG or reverse-bias endurance for NBG); (2) tandem-level protocols that enforce spectral consistency, validate dynamic current matching and simulate multiple stresses; and (3) module-level standards under outdoor conditions that ensure the robustness of encapsulation and the durability of interconnects. Proposing a specialized ISOS-Tandem subclass within the ISOS framework, in collaboration with industry partners, could standardize testing procedures and expedite the commercialization of perovskite tandem cells.

Although efficiency improvements in perovskite devices have progressed rapidly, stability analysis remains more complex and time intensive. Strategies such as introducing additives or interface treatments often enhance efficiency but do not always improve stability, highlighting the need for tailored approaches. Advanced in situ tools such as photoelectron microscopy and synchrotron radiation will be critical for providing dynamic, high-resolution insights into perovskite behaviour under operational conditions. Looking ahead, high-throughput research methods integrated with machine learning could redefine stability studies. By enabling large-scale data collection and analysis, these approaches may help to identify molecular candidates that simultaneously enhance stability and efficiency, accelerating material and device optimization.

Challenges and innovations in upscaling production

Most all-perovskite tandem PSCs reported to have high PCEs are based on laboratory-scale small areas ($<1\text{ cm}^2$) in 2T architecture (details in Box 2), which stimulated scaling up to over 1 cm^2 area and even mini-module dimensions of $>10\text{ cm}^2$ (Fig. 3a). In this section, we focus on the upscaling of 2T all-perovskite tandem devices. Improving the PCEs of large-area all-perovskite tandem devices by scalable methods is challenging, since upscaling from a small area to a large one induces new complexities, such as sequential scalable processing, film homogeneity, interfacial quality, charge transport or collection, and module designs.

Upscaling of tandem solar cells to greater than 1 cm^2

The early stage of upscaling all-perovskite tandem cells achieved a PCE of approximately 24% by employing a novel HTL, $\text{NiO}_x/\text{N4,N4'}$ -di(naphthalen-1-yl)- N4,N4' -bis(4-vinylphenyl)biphenyl-4,4'-diamine, and using formamidinium sulfonic acid to regulate the crystallization of NBG perovskites⁶. In 2023, tandem solar cells with an area of $\sim 1\text{ cm}^2$ benefited from the substantial momentum of research into SAM HTLs in WBG perovskite subcells⁴¹ (Fig. 3b), pushing PCEs up to $>26\%$ with enhanced V_{oc} levels of $>2.1\text{ eV}$. Compared with small-area devices ($\sim 0.1\text{ cm}^2$ in area), tandem solar cells with an area of $\sim 1\text{ cm}^2$ suffer severe PCE loss due to surficial inhomogeneity, especially at the perovskite/ETL interface. Recently, several novel strategies (such as forming 3D/3D or 2D/3D bilayer perovskite heterojunctions (using thermal evaporation) or Dion–Jacobson or Ruddlesden–Popper 2D phases) have achieved uniform passivation, realizing groundbreaking certified PCEs of up to 28.2% on an area of $\sim 1\text{ cm}^2$ (ref. 98). These breakthroughs narrowed the absolute PCE gap to $<1.5\%$ for upscaling from ~ 0.1 to $\sim 1\text{ cm}^2$, providing promising pathways to large-sized module fabrication.

Scalable processing of tandem solar modules

Scaling up perovskite films to large-area module dimensions necessitates moving beyond spin coating to scalable and cost-effective methods, such as slot-die coating, blade coating, thermal evaporation, spray coating and inkjet printing. The currently reported high-efficiency devices are all prepared using the blade-coating method (Fig. 3b). Functional layers, including charge transport layers and TRJs, can be deposited using techniques such as thermal evaporation, blade coating, atomic layer deposition (ALD) and sputtering.

The development of all-perovskite tandem solar modules has lagged behind single-junction solar modules and small-area ($<1\text{ cm}^2$)

tandem solar cells (Fig. 3b). Only a few prototypes have been demonstrated to date^{99–103}, reflecting the significant challenges in transitioning to fully scalable processing for large-area substrates. Sequential deposition of multiple layers (more than ten) and the control of perovskite crystallization for both WBG and NBG layers are critical hurdles. Additionally, transitioning from spin coating to scalable techniques and scaling up from $\sim 1\text{ cm}^2$ to module-sized areas present significant barriers to achieving high performance (Fig. 3b).

Despite these challenges, advancements in module design and scalable coating technologies have enabled an all-perovskite tandem solar mini-module to achieve a certified aperture-area PCE of 24.5%, surpassing the current record of 23.2% for single-junction solar mini-modules. The following sections will explore the key factors driving this performance improvement.

Key parameters of efficiency losses for modules

The module $V_{\text{oc,module}}$ is the sum of the V_{oc} of all subcells. $V_{\text{oc,module}}$ loss mainly originates from the scalable coating quality of the large-area perovskite layer that is dominated by morphological homogeneity, crystal quality, thickness uniformity and interfacial quality. The module short-circuit current ($I_{\text{sc,module}}$) is determined by the smallest I_{sc} of all subcells. $I_{\text{sc,module}}$ loss is mainly associated with the geometric fill factor (that is, the proportion of active area to aperture area; Fig. 3c) and internal series resistance (R_s) of the entire module.

P1 and P3 scribes isolate, whereas P2 scribes connect adjacent subcell electrodes. Accurate control of P2 scribe width is crucial; excessively wide scribes decrease the geometric fill factor and overly narrow ones increase resistive losses due to incomplete layer removal, TCO damage, debris redistribution or metal–halide interactions¹⁰⁴ (Fig. 3c). A thin conformal diffusion barrier layer (that is, $\sim 10\text{ nm}$ ALD- SnO_x) was employed after the P2 scribing process to prevent the interdiffusion between Ag electrodes and I, simultaneously forming good ohmic contact with low vertical resistance at the ITO/ALD- SnO_2 /Ag junction and also improving device stability⁹⁹. This also isolated Ag and PEDOT:PSS in the recombination junction, enhancing shunt resistance. Optimized laser scribing (for example, pulse energy, frequency and speed) ensures uniform ablation and complete layer removal and avoids TCO damage. Additionally, air exposure degradation, particularly in NBG perovskites, must be addressed during module fabrication.

Control of perovskite crystallization in scalable processing

Perovskite film quality and homogeneity are controlled by crystallization kinetics that involves a classical theory of nucleation and crystal growth^{105–113} (Fig. 3d). In general, two pathways are involved in the first step: homogeneous nucleation (that is, when nuclei are generated within the precursor without any preferential sites); and heterogeneous nucleation (that is, when nuclei preferentially form in specific regions, such as on the substrate surface, at defect sites or around impurities)¹¹⁴.

One of the critical challenges is the inherent uncontrollable rapid crystallization of Sn-based perovskites that induces complexity and difficulty in the blade-coating process¹¹⁵. The higher Lewis acidity of Sn^{2+} and lower solubility of SnI_2 (in a dimethylformamide (DMF) or dimethyl sulfoxide (DMSO) solvent system) make the crystallization conditions more stringent and induce a narrow processing window. For example, Sn-based perovskites are capable of crystallizing at room temperature, unlike Pb-based perovskites, which require post-annealing. In addition, the difference in dynamic crystallization kinetics and thermodynamic stability creates competition between Sn and Pb, resulting in an inhomogeneous distribution throughout the perovskite film (for example, imbalanced stoichiometry of Sn and Pb). Therefore, the uncontrollable properties of rapid crystallization generally lead to poor homogeneity, increased defect density and low crystalline quality. This also explains why the scalable processing of NBG perovskite films is currently a key production bottleneck for the advancement of all-perovskite solar modules (Fig. 3e). A moderate nitrogen flow

BOX 2

Performance evolution of all-perovskite tandem solar cells and modules

The box figure illustrates the evolution of all-perovskite tandem solar cells and modules, focusing on key advancements in PCEs and stability from 2016–2025. The development can be categorized into three distinct phases.

Early studies on all-perovskite tandem solar cells focused on finding suitable TRJ architectures and improving the crystallization of NBG subcells. Initial designs used sputtered ITO as the TRJ and ALD-SnO_x as a buffer layer, achieving a PCE of 17.0%¹³⁴. Subsequent advancements came with incorporating the evaporated MoO_x into Ag/ITO interfaces and Cl[−] into NBG perovskites, increasing PCEs to over 21%¹²⁸. Later, ALD-deposited SnO₂ layers paired with ultrathin Au nanoclusters created efficient TRJs, simplifying fabrication and enabling certified efficiencies of up to 24.8% in 2019 (ref. 5). This TRJ design has since become a mainstream choice for high-efficiency tandem architecture. However, at this stage, the operational stability (T_{90}) of tandem devices remained below 500 h (ref. 5).

From 2020, efforts shifted to enhancing the crystallinity and reducing the defects in the perovskite absorber layer of each subcell to boost tandem performance. Key innovations included incorporating 4-trifluoromethyl-phenylammonium into Pb–Sn perovskite precursors, extending the carrier diffusion length and

achieving a certified tandem efficiency of 26.4% with a T_{90} stability of 600 h (ref. 11). Gentle gas quenching methods controlled mixed-halide WBG crystallization, yielding a V_{OC} of 1.33 V, a tandem efficiency of 27.1% and a T_{80} stability of 1,825 h (ref. 19). For tandem modules, the optimized WBG compositions enabled large-area fabrication with a module efficiency of 22.5% and a T_{80} stability of 500 h (ref. 99).

From 2022, research focused on interfaces between perovskites and their respective HTLs or ETLs. These advances led to a reduction in surface defects and improved energy-level alignment, contributing to improvements in both efficiency and stability. For WBG subcells, diammonium molecules were used to enhance surface states, achieving a uniform surface potential and a tandem efficiency of 27.4%¹²⁷. Depositing a lead halide WBG perovskite layer on NBG subcells has created an advanced 3D/3D heterojunction that reduces non-radiative recombination and improves charge extraction, enabling tandem efficiencies exceeding 28% with a T_{90} stability approaching 1,000 h (ref. 25). Tandem modules also experienced significant progress, achieving efficiencies of up to ~25% through advanced crystallization of NBG perovskite and enhanced passivation at the buried interface, achieving a T_{80} stability of beyond 600 h (ref. 100).



Box Fig. 2 a, Timeline illustrating the progression of efficiency and stability in all-perovskite tandem cells and modules^{5,7,10–12,19,25,26,30,34,38,42,56,83,89,99–101,103,127–129,134,136,137}. **b**, Evolution of typical and universal TRJs in all-perovskite tandem cells. BCP, bathocuproine; PCBM, [6,6]-phenyl-C₆₁-butyric acid methyl ester. Panel **b** (top) adapted with permission from ref. 134, AAAS. Panel **b** reproduced with permission from: middle, ref. 128,

Springer Nature Limited; bottom, ref. 5, Springer Nature Limited.

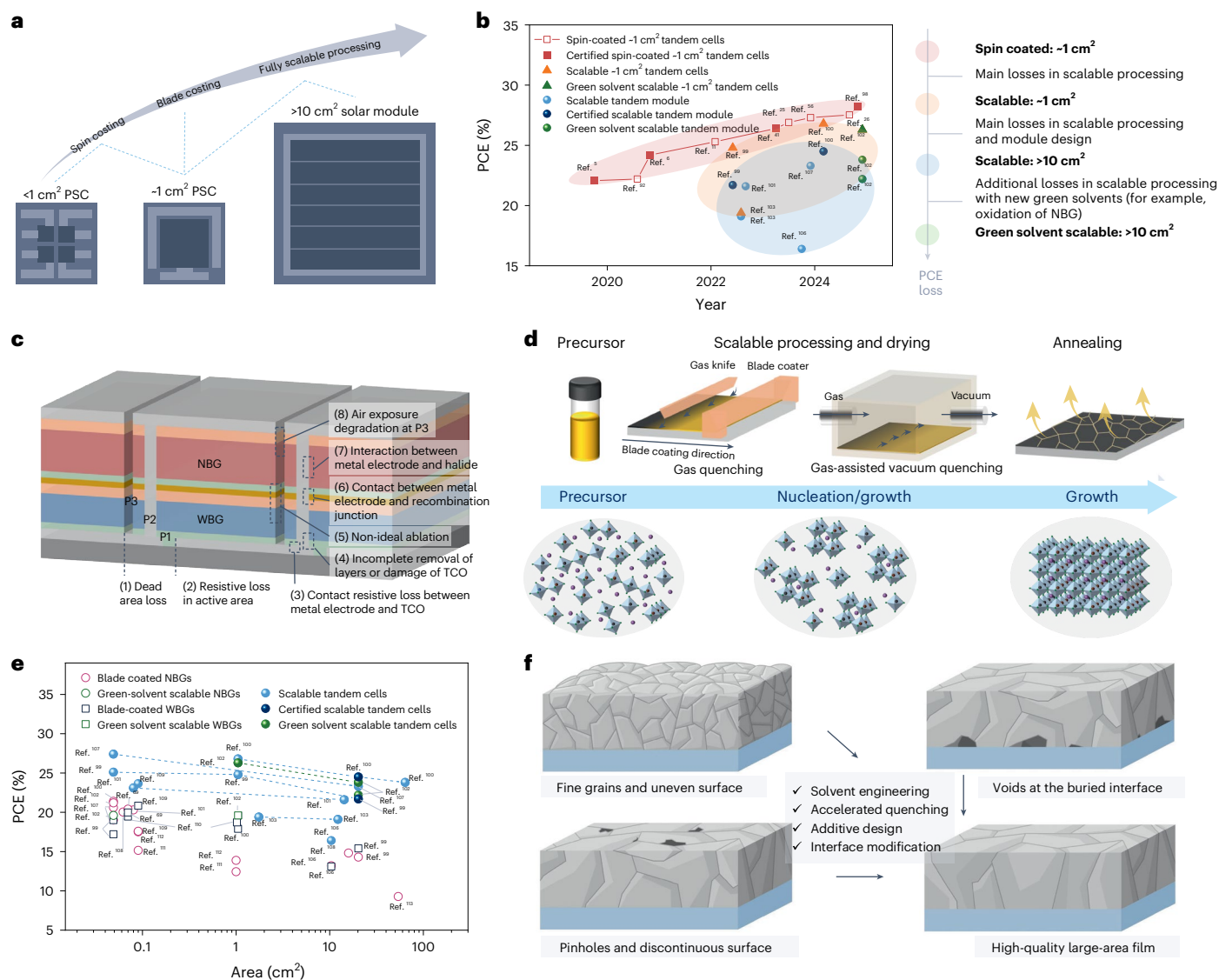


Fig. 3 | Upscaling the fabrication of all-perovskite tandem cells. **a**, Illustration of upscaling from a small area (<1 cm²; spin coating) to large-area (>10 cm²; spin coating, blade coating or fully scalable processing) or module dimensions (>10 cm²; fully scalable processing). **b**, PCE evolution of -1 cm² all-perovskite PSCs^{5,6,11,25,26,41,56,92,98,105} and mini-modules^{99–103,106,107}. **c**, Schematic of all-perovskite solar modules with series interconnection and summarized possibilities of

geometrical losses in module designs. **d**, Schematic of the scalable processing of large-area perovskite films and key control of crystallization. **e**, PCE progress of single-junction and all-perovskite tandem PSCs or modules based on the blade-coating process^{69,99–103,106–113}. **f**, Schematic of typical crystallization problems and mitigation strategies in large-area perovskite fabrication.

(260 cm³ min⁻¹) was employed for fabricating NBG perovskite film with an area of 12.25 cm², resulting in an accelerated mass transport from the NBG perovskite surface and improved crystallization. It produced a uniform morphology free of defects and pinholes (Fig. 3f), achieving a PCE of 19.1% in scalable mini-modules¹⁰³. To blade coat a thick (~1 μm) NBG perovskite film, the as-fabricated wet film also needs to be thick. However, the trapped DMSO (a solvent with a high boiling point) near the buried interface induces interfacial voids due to the top-to-bottom crystallization direction^{100,101}. To avoid the void formation, a reduced molar ratio of DMSO (with an optimum of 8% DMSO relative to the concentration of Pb and Sn) was utilized in the precursor, resulting in enlarged grain sizes and a void-free buried interface¹⁰¹ (Fig. 3f). This enabled efficient all-perovskite tandem solar modules delivering a PCE of 21.6% over a 14.3 cm² aperture area. It was also found that the wet-tability of NBG perovskites decreases on top of large-area PEDOT:PSS substrates with a decay of the coating time, leading to a non-uniform density of nuclei following uneven solvent extraction. Therefore, it is crucial to extend the processing window of blade coating as this

is a time-demanding step. A novel additive, aminoacetamide hydrochloride, was employed to retard the crystallization as it coordinates strongly with the precursor. The processing window was extended up to 100 s without performance loss due to high homogeneity and the quality of the NBG perovskite films¹⁰⁰ (Fig. 3f). Aminoacetamide hydrochloride was also shown to accumulate at the buried interface with passivating defects. Fully scalable all-perovskite tandem solar modules achieved a certified PCE of 24.5% with a 20.25 cm² aperture area (Fig. 3b). This approach also reached a high PCE of 23.8% with a 64 cm² aperture area (Fig. 3e). Recently, the same group released a record certified PCE of 24.8% over a 64 cm² aperture area¹.

Scalable processing of WBG perovskite films remains challenging. Mixed-halide WBG perovskites exhibit diverse crystallization kinetics, often accompanied by phase heterogeneity issues^{69,99}. Additionally, Cs and Br salts have limited solubility in polar aprotic solvents, such as DMF and DMSO. These constrain the scalable processing of large-area WBG perovskite films for module production. In response, nucleation was controlled by varying the Cs content. The optimized composition

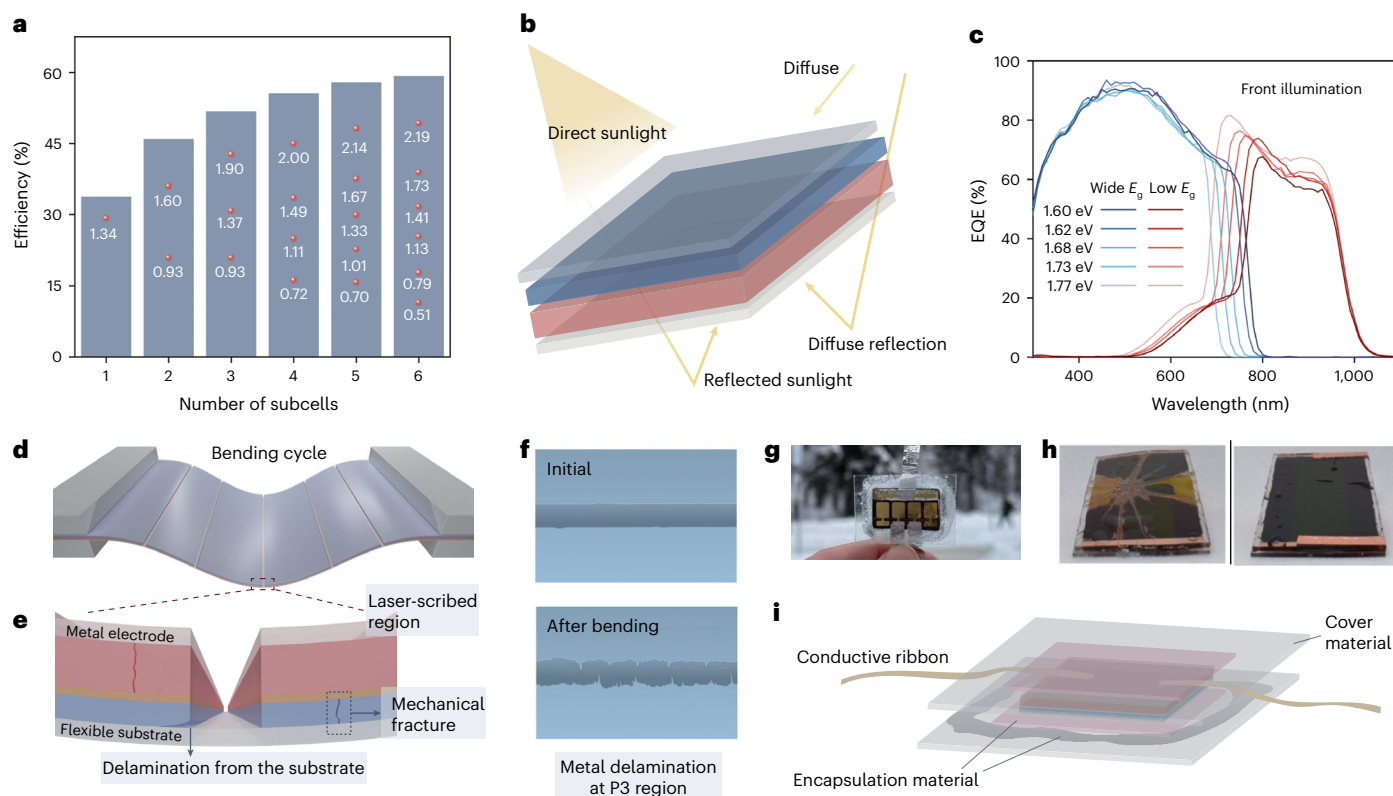


Fig. 4 | Prospects of all-perovskite tandem photovoltaics. **a**, Theoretical efficiency limits of multi-junction solar cells comprising different numbers of subcells. The optimum bandgaps (eV) of the individual cells when in combination are labelled as individual data points. Data derived from ref. 116 based on detailed balance limits. **b**, Schematic of energy harvesting in a bifacial tandem device, including direct sunlight, diffused light due to cloud cover and reflected sunlight from the surrounding environment. **c**, External quantum efficiency (EQE) curves of WBG and NBG subcells in bifacial all-perovskite tandem cells with various top cell bandgaps under front (glass side) illumination. **d**, Schematic of the bending test for a flexible module under bending. **e**, Schematic of mechanical fracture

and delamination in a flexible tandem module after bending tests. **f**, Schematic of metal delamination at the P3 region in a flexible tandem module after bending tests. **g**, Photo of a one-step-coated polydimethylsiloxane-encapsulated device. **h**, Photos of ethylene-vinyl acetate- (left) and shellac-encapsulated (right) perovskite modules after hail impact testing. **i**, Schematic of perovskite device encapsulation. Panels adapted with permission from: **c**, ref. 123 under a Creative Commons license CC BY 4.0; **g**, ref. 125 under a Creative Commons license CC BY 4.0. Panel **h** reproduced with permission from ref. 126 under a Creative Commons license CC BY.

of $\text{Cs}_{0.35}\text{FA}_{0.65}\text{PbI}_{1.8}\text{Br}_{1.2}$ realized enlarged grain sizes, improved film homogeneity and reduced phase segregation⁹⁹. A certified PCE of 21.7% was achieved in mini-modules (Fig. 3e). Some additives (such as methylammonium thiocyanate¹⁰⁹ and glycine amide hydrochloride⁶⁸) were also employed to regulate crystallization in the blade-coating process. Very recently, green solvent scalable fabrication was developed in all-perovskite tandem solar modules. To replace DMF, a green solvent system comprising DMSO/acetonitrile/ethyl alcohol was developed to extend the processing window and avoid void formation at the buried interface¹⁰² (Fig. 3f). Utilizing green solvents in WBG perovskite precursors achieved a high PCE of 23.8% in tandem modules. The green solvent system also demonstrated upscaling capability in NBG perovskite fabrication. As a result, a mini-module based on fully green solvents for the processing of both WBG and NBG perovskites realized a PCE of 22.2% (Fig. 3e). The PCE gap (Fig. 3b) was attributed to the Sn^{2+} oxidation caused by DMSO, as a higher ratio of DMSO was used in the green solvent system⁶⁴. This achievement opens the door for the environmentally friendly production of all-perovskite tandem modules.

Perspectives and outlooks

As highlighted in the previous section, all-perovskite tandem solar cells have garnered significant research interest in recent years. This section will discuss potential directions for future exploration, key challenges and outlooks.

Multi-junction configurations for enhanced efficiency

Multi-junction architectures represent transformative strategies for surpassing the efficiency limits of tandem devices by harnessing a broader portion of the solar spectrum^{116,117}. The theoretical efficiency limit increases towards more junctions in combination with optimum bandgaps (Fig. 4a). Unlike the high-cost III–V multi-junction cells, perovskite-based multi-junction solar cells offer a cost-effective and high-performance alternative for terrestrial applications. However, additional subcells introduce new complexity and challenges, such as sequential processing, device architectures, the interaction between the multi-layer stacks, the choice of ultra-NBG perovskites, stability and upscaling. It is noteworthy that the lowest perovskite bandgap (down to ~1.2 eV) impedes the development of all-perovskite multi-junction photovoltaics (for example, at least two subcells need optimum ultra-NBGs of <1.15 eV for four or more junctions). Very recently, a new benchmark for multi-junction photovoltaics was achieved by enhancing Pb–Sn perovskite films with amino acid salts, which synergistically improved the film quality and optoelectronic properties, leading to record PCEs of 28.7% for triple-junction devices and 27.9% for quadruple-junction devices¹¹⁸. Despite these advances, significant gaps remain in reaching the practical efficiency potential of 37%¹¹⁷ (for ideal bandgaps of 1.2, 1.5 and 2.0 eV).

These differences between theoretical and experimental efficiencies are primarily attributed to the performance limitations in WBG (2.0 eV) top cells, where a high Br content complicates the film quality.

Due to the versatility of WBG cells, the optimization of all-perovskite multi-junction devices often aligns with advancements in perovskite/Si and perovskite/organic multi-junction technologies. Emerging strategies to overcome these challenges include the use of halogenated diammonium salts and propane-1,3-diiodide as additives to enhance halide homogeneity. Compositional engineering, such as replacing Cs^+ with Rb^+ at the A site and introducing halogenocyanate (OCN^-) at the X site, has shown promise in alleviating lattice strain^{119,120}. Realizing the full potential of multi-junction architectures will require continued optimization of subcell performance, complemented by advanced light management techniques, including anti-reflective coatings and photonic structures, to ensure optimal photon utilization and current matching.

Real-world optimization with structure design

The real-world performance of all-perovskite tandem photovoltaics depends on adapting to variables such as tilt angles, tracking methods and climatic factors, including solar spectra and humidity¹²¹. Considering albedo irradiation under realistic irradiation conditions, the bifacial tandem photovoltaic retains a higher energy yield gain compared with the monofacial photovoltaic¹²². For bifacial tandem modules, effective light management becomes critical to achieving current matching and maximizing energy capture from both direct and diffuse light (Fig. 4b). Advanced strategies, such as bandgap tuning¹²³ (Fig. 4c), angular response optimization and thermal management systems, are being developed to address these challenges.

A key hurdle under field test conditions is current mismatch, where varying angles of incidence and spectral shifts disrupt the subcell photocurrent balance. Unlike the controlled conditions of standard test protocols, field test conditions expose devices to dynamic thermal effects, wind and humidity, which can alter perovskite bandgaps and exacerbate mismatch. Despite these challenges, tandem photovoltaics have demonstrated the ability to outperform single-junction devices under diffuse light and non-standard spectra, highlighting their potential for superior field performance.

All-perovskite tandem flexible modules, combining high efficiency with mechanical adaptability, are well suited for wearable electronics, curved surfaces and building-integrated photovoltaics (Fig. 4d). However, achieving long-term operational stability under mechanical stress, environmental exposure and thermal cycling remains a formidable challenge (Fig. 4e). Studies have shown that replacing NiO_x with NiO_x/SAM transport layers significantly enhances mechanical resilience¹⁰⁵. Further progress will require the optimization of intermediate TRJ structures to balance electrical performance with bending flexibility, as well as mitigating delamination and thermal effects induced by laser scribing. For instance, bending-induced metal delamination along P3 grooves remains a critical issue, increasing the risk of short circuits¹²⁴ (Fig. 4f). Addressing these limitations is essential for advancing the reliability and scalability of flexible tandem modules.

Environmental life cycle enhancement

The environmental footprint of all-perovskite tandem solar cells is a critical consideration for their large-scale adoption. Future industrialization should prioritize transitioning to fully green solvent- or evaporation-based fabrication methods. Encapsulation plays a dual role: it ensures long-term device stability and serves as a vital barrier against Pb leakage during operation. Current encapsulation techniques include single-layer coatings, such as epoxy resin or silicone cured by exposure to ultraviolet light¹²⁵ (Fig. 4g), and multi-step hot pressing using thermoplastic polymers¹²⁶ (Fig. 4h,i). Among these, multi-step hot pressing is widely employed in industrial settings due to its robust protective capabilities, although it necessitates further innovation to enhance efficiency and flexibility while minimizing efficiency losses during the heating process. Future advancements should focus on developing high-performance multi-layer barrier films, advanced adhesive interfaces and innovative flexible encapsulation designs.

Despite the many challenges that remain in realizing the full potential of all-perovskite multi-junction devices, the sustained effort and rapid progress in perovskite research offer a promising outlook. We strongly believe that all-perovskite tandem cells present the ultimate embodiment of perovskite technology, paving the way for the development of highly efficient and cost-effective thin-film modules, and ultimately driving their large-scale manufacture and deployment.

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Author contributions

J.W., H.H., C.C. and D.P.M. jointly contributed to the conceptualization, literature analysis, figure preparation, and manuscript writing. R.L., K.X. and Y.L. compiled the data for the article. H.T., H.J.S., U.W.P. and J.T. reviewed and/or edited the manuscript before submission.

Competing interests

H.T. is the founder, Chief Scientific Officer and Chairman of Renshine Solar, a company that is commercializing perovskite photovoltaics. H.J.S. is the co-founder and Chief Scientific Officer of Oxford PV. The other authors declare no competing interests.

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